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Generative AI for wearable information fusion: Evidence mapping the reality gap in activity recognition and health monitoring

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ABSTRACT

Generative models are being used to address data scarcity, modality gaps, and semantic grounding in wearable Human Activity Recognition (HAR) and Health Monitoring (HM), but deployment evidence remains limited. This focused evidence-map review examines Scopus-indexed Q1/Q2 journal literature on generative Artificial Intelligence (AI) for wearable information fusion, focusing on datasets, validation practices, and deployment claims. A PRISMA-guided Scopus search, followed by Q1/Q2 journal filtering and backward reference checking, yielded a frozen 2023–2025 corpus of 22 studies: 17 primary, 4 reviews/surveys, and 1 framework paper. In this corpus, Generative Adversarial Networks (GANs) are the most frequent generative family for augmentation, class balancing, and biosignal synthesis. Large Language Models (LLMs) appear mainly as semantic or interface layers for activity description, pseudo-label assessment, knowledge retrieval, and decision-use prototypes, rather than as clinically validated systems. The primary studies show a consistent reality gap: downstream classification or recognition gains on benchmark tasks are commonly reported, but physiological plausibility, privacy evaluation, clinical or contextual grounding, and deployment-level generalization remain sparse or indirect. We map each study to information-fusion roles, assess dataset representativeness and alignment with intended deployment settings, and summarize validation coverage through a qualitative matrix. The corpus supports technical progress, but the evidence is narrower than deployment-oriented language sometimes implies. Stronger deployment claims require richer longitudinal wearable datasets, validation beyond benchmark utility, explicit privacy reporting, and clinical claims matched to the available evidence.

1. Introduction

Wearable sensing is used in both Human Activity Recognition (HAR) and Health Monitoring (HM): accelerometers, gyroscopes, pressure sensors, Electrocardiogram (ECG), Photoplethysmogram (PPG), and related on-body sensors are used to detect daily activities, monitor physiological status, estimate stress and sleep, and support remote care pathways [1,2]. As proposed systems are discussed for persistent real-world monitoring, the limiting factor is often not the classifier or sequence model but the availability of datasets that are sufficiently diverse, representative, and well annotated for the intended use case.

Generative AI is often proposed as a response to that data bottleneck. Across wearable HAR and HM, generative models are used to augment minority classes, translate between modalities, synthesize biosignals, and improve performance under scarce labels [3–6]. LLMs are also entering this area, but mainly as semantic or interface-oriented components rather than as autonomous clinical systems [7–10]. That shift makes datasets, validation protocols, and deployment assumptions necessary for assessing progress.

The studies at this intersection combine, translate, or semantically enrich heterogeneous evidence streams: inertial and physiological sensors, pressure and motion representations, generated biosignals, natural-language descriptions, retrieved medical knowledge, federated wearable data, and prototype decision-use interfaces. The key assessment is whether benchmark gains also support trustworthy data-level, modality-level, knowledge-level, or decision-level fusion under the incomplete and weakly grounded data conditions common in wearable health systems. The fusion-role labels used below are a corpus-specific operational

Abbreviations: AI, Artificial Intelligence; GAN, Generative Adversarial Network; HAR, HumanActivity Recognition; HM, Health Monitoring; LLM, Large Language Model; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

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coding scheme for interpreting the included papers; they are not intended as a high-recall taxonomy of Information Fusion research or as an additional eligibility screen.

Broader reviews already frame adjacent parts of this area. Kaseris et al. [1] survey deep learning methods in HAR, Amiri et al. [11] review IoT-based bio- and medical informatics, Nasarian et al. [12] review interpretable Machine Learning (ML)/Explainable Artificial Intelligence (XAI) for clinician-AI collaboration in healthcare, or Wang et al. [13] examine smart glasses in digital health. These reviews provide context, but they leave a narrower question unresolved: how journal literature at the intersection of wearable sensing, HAR/HM, generative models, and datasets validates its claims, handles data limitations, and supports arguments about real-world usefulness.

We address that question through a focused evidence-map review of Scopus-indexed Q1/Q2 journal literature targeted to the 2023–2025 period and anchored to a frozen source-of-record retrieval archived on 3 December 2025. The scope is intentionally narrow: wearable or closely sensor-derived HAR/HM, explicit use of generative modeling or LLMs, and substantive engagement with datasets, data generation, or data-driven evaluation. This design prioritizes analytical depth over broad recall. The resulting claims should therefore be read as a synthesis of evidence from that frozen Scopus-indexed Q1/Q2 corpus, not as a census of all interdisciplinary work in the area.

Our central argument is that, within this frozen corpus, the literature more often optimizes technical performance than it validates deployment-relevant reliability. Many papers report distributional fidelity or downstream classification gains, but far fewer examine physiological plausibility, clinically grounded relevance, population transfer, or privacy risk. We refer to that reporting asymmetry as the *reality gap*: synthetic or semantically aligned outputs may improve benchmark metrics while leaving the conditions for trustworthy health deployment only weakly evaluated.

This paper makes four main contributions:

- A fusion-role evidence map that assigns every included paper to data synthesis/augmentation, modality translation, semantic/language grounding, knowledge/retrieval fusion, federated/distributed fusion, contextual review, and/or decision-use prototype roles while identifying validation gaps for those roles.
- A study-type-stratified synthesis that separates primary studies, reviews/surveys, and framework papers rather than treating the evidence base as methodologically uniform.
- An analysis of how dataset representativeness, ecological validity, longitudinal coverage, annotation richness, and clinical fit constrain claims about wearable information fusion.
- A qualitative validation matrix that shows where the primary studies do and do not report distributional fidelity, downstream utility, generalization, physiological plausibility, privacy mechanisms or evaluation, and clinical/context grounding.

Section 2 defines the review design and screening workflow. Section 3 presents the study-type-stratified synthesis of the included studies. Section 4 examines dataset constraints through a deployment-oriented lens. Section 5 interprets the reality gap and its implications for evaluation and reporting. Section 6 concludes with research priorities.

2. Methodology

We conducted a focused evidence-map review designed to support critical synthesis rather than full coverage of the field. The aim was to assemble a transparent journal evidence base for assessing datasets, validation practices, and deployment claims in generative wearable information fusion. Reporting follows PRISMA-guided review practice and recent guidance on search transparency [14–17], but the design should not be interpreted as a field-wide systematic review. No external prospective protocol was registered; the Supplementary Material documents the implemented review protocol details, archived search expression, eligibility rules, screening count summary, extraction schema, validation dimensions, and dataset and resource catalog.

2.1. Search process and screening strategy

The formal source of record for the review was Scopus. It provided consistent indexing, journal metadata, and a stable query environment for the targeted scope. The query snapshot used for the source-of-record retrieval was archived on 3 December 2025, and full-text assessment continued into early 2026. The review targeted publications from 2023 to 2025, but the effective corpus was intentionally frozen at that archived retrieval date, apart from one additional paper admitted through documented backward reference checking.

The search terms targeted wearable or sensor-derived HAR and HM studies that explicitly engaged generative models, LLMs, or dataset-related questions. Terms such as *Generative*, *Language Model*, and *LLM* captured generative or language-enabled methods, while *dataset* and *database* kept the search centered on studies in which datasets, data generation, or data-centric evaluation were explicit. The term *healthcare* was used as a broad health-monitoring term, not as a high-recall synonym set for all health-monitoring concepts. The Boolean string used for the Scopus search was:

```
("Human Activity Recognition" OR HAR OR
"healthcare") AND ("wearable*" OR sensor*) AND
("Generative" OR "Language Model" OR LLM) AND
(dataset OR database)
```

The query was implemented with Scopus advanced search and translated into the following Scopus expression:

```
TITLE-ABS-KEY ( ("Human Activity Recognition" OR
HAR OR "healthcare") AND ("wearable*" OR sensor*)
AND ("Generative" OR "Language Model" OR LLM)
AND ("dataset" OR "database") ) AND PUBYEAR >
2022 AND PUBYEAR < 2026 AND (LIMIT-TO (SUBJAREA,
"COMP")) AND (LIMIT-TO (DOCTYPE, "ar")) AND
(LIMIT-TO (LANGUAGE, "English"))
```

The `TITLE-ABS-KEY` field searched titles, abstracts, and author-defined keywords. Additional limits restricted the review to the Computer Science subject area, English-language journal articles, and the target publication window.

The query should be read as data-centric and generative-wearable centered. It did not use a high-recall information-fusion or health-monitoring keyword strategy; instead, the included corpus is analyzed for fusion relevance after selection. This means that fusion-relevant work lacking any of the selected dataset, generative-method, wearable/sensor, *healthcare*, or HAR terms may fall outside the preserved corpus by design.

Executing the query in Scopus yielded 47 records. The single-database retrieval did not require a duplicate-removal count adjustment. One additional article was added through backward reference checking after being checked against the frozen Scopus set, resulting in 48 records entering screening. Title/abstract review and journal-quartile screening excluded 23 records, leaving 25 papers for full-text assessment. Three further papers were excluded at full-text stage because they did not meet the final scope: static medical-image encryption (n = 1), text-only Natural Language Processing (NLP)/QABot without wearable grounding (n = 1), and non-wearable data fit (n = 1). The final included set contains 22 papers and is summarized in Fig. 1. The included set therefore reflects a frozen source-of-record corpus rather than a claim of full end-of-year capture. Potentially relevant papers published or indexed after 3 December 2025 were not added, apart from the single backward-snowballing addition already noted in the screening count summary.

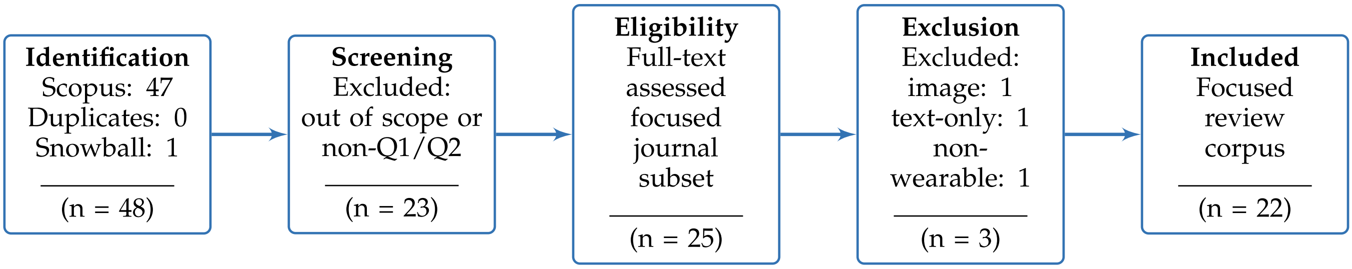

Fig. 1. PRISMA-guided aggregate flow diagram for the focused evidence-map review. The figure summarizes counts only; record-level excluded-title and DOI traceability is not preserved in the manuscript package.

The inclusion criteria were: (i) publication between 2023 and 2025; (ii) English-language journal article indexed in Scopus; (iii) Q1 or Q2 journal status recorded during screening; (iv) substantive relevance to wearable or sensor-derived HAR and/or HM; and (v) explicit use of generative modeling, LLMs, or data-generation/data-centric methodology. The Q1/Q2 filter was retained as a journal-level scope restriction, not as a study-quality score. This restriction reduces recall but concentrates the review on a predefined journal subset with Scopus-indexed quartile metadata. Some relevant interdisciplinary or lower-quartile work may be omitted, so the review prioritizes focused depth over exhaustive breadth.

Borderline cases were resolved conservatively at full-text stage. Papers were retained only when wearable or closely sensor-derived data played a substantive role in the method, evaluation, or literature synthesis, and when the generative or language-model component was central rather than incidental. After full-text assessment, each included paper was labeled by study type: primary empirical study, review/survey, or framework paper. This distinction prevents contextual review articles and empirical validation studies from being interpreted as equivalent evidence.

The first author led screening, extraction, and narrative synthesis. Full dual independent screening of the entire set was not feasible within the available authorship resources. Co-authors therefore spot-checked sampled and borderline inclusion decisions, and disagreements were resolved by discussion. The workflow provides an audit step, but it should not be interpreted as full inter-reviewer agreement reporting.

For the validation matrix, the $Y/P/N$ codes were assigned using the operational definitions reported in the table caption and Supplementary Material. Y denotes direct evaluation of the dimension in the source study, P denotes partial, indirect, context-only, or proxy evidence, and N denotes evidence not explicitly reported in the preserved source record. Because the screening and coding workflow was not implemented as full independent dual coding, no Cohen's kappa or other inter-rater reliability statistic was calculated or claimed. The reproducibility safeguards were predefined extraction fields, explicit validation-dimension definitions, co-author spot checks of sampled and borderline decisions, consensus resolution, and per-study coding rationales in the Supplementary Material.

We also did not perform formal risk-of-bias or study-quality scoring. Given the focused evidence-map design and the heterogeneity of tasks, datasets, and evaluation settings, the synthesis instead relies on structured evidence mapping, study-type stratification, and the validation matrix to state evidentiary boundaries at the corpus level.

No formal meta-analysis was attempted. The included studies are heterogeneous in task definitions, sensing modalities, datasets, evaluation metrics, and validation settings, making quantitative pooling inappropriate. Instead, we use structured evidence-map-style tables [18] to compare study types, datasets, fusion roles, and validation dimensions across the review. Supplementary Material documents the archived Scopus search expression, screening count summary, study-type classification, final-corpus traceability table, extraction schema, validation dimensions, and dataset and resource catalog underlying this focused evidence-map review.

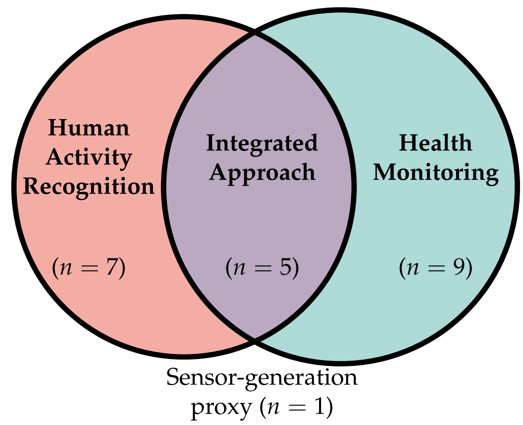

Fig. 2. Distribution of 22 papers across HAR, HM, integrated, and proxy categories.

3. Selected studies and cross-study synthesis

The review includes 22 papers identified through the workflow described in Section 2. As summarized in Fig. 2, seven center on HAR (31.8%), nine on HM (40.9%), five address integrated HAR + HM problems (22.7%), and one is a generic sensor-generation proxy included for health-monitoring applicability (4.5%). The evidence base is methodologically mixed: 17 papers are primary empirical studies (77.3%), 4 are reviews or surveys (18.2%), and 1 is a framework paper (4.5%). We therefore treat the literature as a structured evidence map rather than as a homogeneous set of comparable experiments.

3.1. Positioning against adjacent reviews

As noted in the Introduction, the review papers in the corpus provide adjacent context rather than direct evidence about generative wearable validation [1,11–13]. Table 3 therefore keeps them analytically separate from the primary studies. This paper contributes a corpus-bounded evidence map that relates Scopus-indexed Q1/Q2 generative wearable studies to information-fusion roles and to the validation practices reported for those roles.

3.2. Study-type stratification

Table 1 operationalizes the information-fusion axis used in the synthesis. The labels are non-exclusive: a study may synthesize data, translate modalities, ground sensor evidence in language, connect retrieved knowledge, learn across distributed clients, and feed a prototype decision-use interface. This is a corpus-specific coding scheme for

Table 1
Non-exclusive fusion-role taxonomy used to interpret the included corpus. Role codes are used in Table 2. Labels describe the role played by generative or language-enabled methods in the reviewed studies; they are not a new eligibility screen.

Fusion role	Operational meaning in this review	Representative included studies	Validation implication
Data synthesis / augmentation (DS)	Generated samples, noise, or minority-class examples are fused into model training or evaluation pipelines.	Chen et al. [3], Yang et al. [4], Sohn et al. [5], Aleroud et al. [20], Zhang et al. [21], Jeong et al. [22], Djemaa et al. [23]	Utility gains should be paired with signal plausibility, transfer, and privacy checks.
Modality translation (MT)	One evidence stream is translated into another, such as motion-to-IMU, pressure-to-language, text-to-pressure, or ECG-to-PPG.	Sohn et al. [5], Leng et al. [7], Ray et al. [8]	Translation quality should be evaluated beyond benchmark accuracy, especially for alignment and physiological meaning.
Semantic or language grounding (SL)	Sensor or activity evidence is linked to natural-language descriptions, symbolic activity models, or language-model classifiers.	Leng et al. [7], Ray et al. [8], Djemaa et al. [23], Oğul [24]	Semantic usefulness should not be interpreted as clinical validation unless clinical endpoints are tested.
Knowledge / retrieval fusion (KR)	Wearable-derived outputs are combined with external medical knowledge, dialogue resources, or health-management rules.	Yang et al. [9], Li et al. [19]	Knowledge-grounded interfaces remain prototype evidence unless prospective decision performance is evaluated.
Federated or distributed fusion (FD)	Wearable evidence is learned across distributed clients, label sources, or pseudo-labeling stages.	Bukit et al. [10]	Privacy-oriented design should be separated from measured leakage risk or formal guarantees.
Decision-use endpoint / prototype interface (DU)	Generated, translated, semantic, retrieved, or modeled temporal evidence supports a stated monitoring, anomaly-detection, diagnosis-oriented, or health-management endpoint, without implying validated clinical decision support.	Yang et al. [9], Li et al. [19], Jeong et al. [22], Bacciu et al. [25], Errafik et al. [26], Jin et al. [27]	Intended use, target population, workflow fit, safety limits, and decision endpoints should be explicit.

interpreting the included papers, not a high-recall Information Fusion taxonomy or a new eligibility filter. Table 2 applies those labels to every included paper and states the fusion-role mapping directly. Role assignments are based on the preserved model family, claimed contribution, validation setting, and synthesis description, and they do not assert implementation maturity. Among the 17 primary studies, data synthesis or augmentation is the most frequent role (10/17, 58.8%), followed by semantic or language grounding (6/17, 35.3%), decision-use endpoints or prototype interfaces (5/17, 29.4%), modality translation (3/17, 17.6%), knowledge or retrieval fusion (1/17, 5.9%), and federated or distributed fusion (1/17, 5.9%).

Table 3 summarizes the evidence base. Rows are grouped by study type and domain, and the primary-study sequence is mirrored in the validation matrix. The table separates three evidence classes. Primary studies dominate the corpus numerically and provide most of the empirical evidence on augmentation, sensor translation, privacy-aware design, semantic enrichment, and prototype decision use. Reviews and surveys provide broader context on HAR, digital health, and interpretability, but they do not directly validate generative wearable systems. The single framework paper by Li et al. [19] is conceptually useful yet empirically limited. That distinction matters because claims about maturity or clinical utility should rest primarily on the primary studies.

3.3. Cross-study patterns

Four cross-study patterns inform the discussion that follows. First, GANs are the most frequent single generative family in the review [3–5,23]. They are used chiefly for class balancing, biosignal augmentation, and synthetic sample generation rather than for open-ended sequence modeling. Variational Autoencoders (VAEs), autoencoders, and related architectures appear mainly in domain adaptation or feature-learning roles [6,26,28]. Diffusion-model primary studies are not represented in the preserved Scopus Q1/Q2 corpus, so the findings should not be generalized to diffusion-based wearable synthesis. By contrast, LLMs are typically used as semantic support layers for text generation, natural-language interfaces, pseudo-label evaluation, or knowledge retrieval rather than as stand-alone clinical systems [7–10]. From an information-fusion perspective, these studies mostly use generative methods to fill sparse data regions, translate between signal and semantic spaces, or connect sensor streams with external knowledge. They do not validate end-to-end clinical fusion pipelines.

Second, validation remains concentrated on benchmark performance. Most primary papers report downstream utility as accuracy, macro F1, anomaly-detection performance, or improvement over a baseline on one or more public datasets. Explicit evaluation of generalization or transfer beyond bounded benchmarks, physiological plausibility, prospective deployment behavior, or clinical end-point relevance is far less common. Even when papers address healthcare-oriented tasks, validation often remains anchored to proxy machine-learning outcomes rather than clinical decision quality or longitudinal monitoring performance.

Third, the comparison between HAR and HM is mainly a difference in validation target rather than evidentiary maturity. HAR studies more often use centralized public benchmark evaluation and report classification gains, whereas HM studies more often invoke diagnosis-oriented, physiological, pediatric, mood, or health-management contexts. The latter contexts make deployment implications more visible, but they do not by themselves provide external clinical validation. Similarly, the only federated or distributed primary study in the corpus, Bukit et al. [10], remains benchmark-bounded despite its privacy-oriented architecture, while the remaining primary studies rely on centralized or study-local evaluation settings.

Fourth, the review and framework papers largely reinforce rather than overturn these observations, emphasizing multimodal integration, privacy, interpretability, and dataset quality [1,11–13,19]. They help situate the field, but they do not substitute for empirical validation.

The included papers differ substantially in sensing modality, task definition, cohort design, benchmark choice, and evaluation metric, making quantitative pooling inappropriate. The remainder of the paper therefore relies on structured qualitative synthesis and explicit gap mapping.

4. Dataset evidence and critical appraisal

Datasets determine what the reported gains can plausibly mean in this literature. Table 4 summarizes the dataset families and constraints that most strongly shape generative wearable information-fusion research: representativeness, ecological validity, longitudinal

Table 2
Per-study fusion-role map for the included corpus. Role codes are defined in Table 1; labels are non-exclusive and describe how the included paper is used in the evidence map. Contextual review papers are not assigned primary empirical fusion-role codes.

Included paper	Evidence class	Fusion role code(s)	Assignment basis
Chen et al. [3]	Primary	DS	Class-conditioned multimodal synthetic wearable HAR augmentation.
Aleroud et al. [20]	Primary	DS	GAN/microaggregation alteration of HAR data with privacy-utility evaluation.
Leng et al. [7]	Primary	MT; SL	Language-to-motion/IMU transfer and text-aligned activity descriptions.
Ye and Wang [6]	Primary	DS	Generative domain adaptation for cross-user HAR.
Oğul [24]	Primary	SL	Symbolic/language-like action sequence modeling.
Ray et al. [8]	Primary	MT; SL	Reciprocal pressure-language generation.
Zhang et al. [21]	Primary	DS	Differentiable generative/noise augmentation for sensor HAR.
Bacciu et al. [25]	Primary	DU	Wearable mood-event prediction endpoint.
Romanelli and Martinelli [28]	Primary	DS	Synthetic sensor measurement/noise generation.
Sohn et al. [5]	Primary	MT; DS	ECG-to-PPG biosignal translation used for AF-classifier augmentation.
Yang et al. [4]	Primary	DS	Biosignal/time-series augmentation for scarce classes.
Yang et al. [9]	Primary	SL; KR; DU	Wearable inputs, retrieved medical knowledge, and dialogue prototype.
Jeong et al. [22]	Primary	DS; DU	Synthetic minority sampling for childhood weight-management prediction.
Bukit et al. [10]	Primary	FD; SL	Federated wearable learning with LLM pseudo-label filtering.
Djemaa et al. [23]	Primary	DS; SL	Controllable GAN augmentation with LLM-based classification.
Errafik et al. [26]	Primary	DS; DU	Augmentation pipeline for elderly-monitoring HAR endpoint.
Jin et al. [27]	Primary	DU	Abnormal-behavior detection endpoint from multi-sensor time series.
Amiri et al. [11]	Review	Contextual review; no primary empirical role code	Contextual review synthesis only; no primary empirical fusion-role code.
Nasarian et al. [12]	Review	Contextual review; no primary empirical role code	Contextual review synthesis only; no primary empirical fusion-role code.
Wang et al. [13]	Review	Contextual review; no primary empirical role code	Contextual review synthesis only; no primary empirical fusion-role code.
Kaseris et al. [1]	Survey	Contextual survey; no primary empirical role code	Contextual survey synthesis only; no primary empirical fusion-role code.
Li et al. [19]	Framework	KR; DU	Framework-level knowledge/rule-guided LLM generation and health-management decision-use concept.

coverage, annotation richness, and clinical fit. The detailed dataset and resource catalog remains in the Supplementary Material, but the available manuscript package does not provide consistent participant-demographic, free-living, label-source, access/license, or clinical-target metadata for every resource. The dataset critique is therefore qualitative and corpus-bounded.

4.1. Benchmark suitability and representativeness

Most included HAR studies still depend on canonical benchmarks such as UCI-HAR [29], PAMAP2 [30], WISDM [31], Opportunity [32], and RealWorld [33]. These datasets were built primarily for activity-recognition benchmarking, not for clinically grounded deployment. They remain useful for controlled comparison, but many are dominated by healthy adults, short recording sessions, and highly scripted activities. That matters when papers extend their claims toward elderly monitoring, pediatric monitoring, fall-risk assessment, or chronic-disease support. The studies by [22] and [26], which move closer to vulnerable or clinically relevant populations, remain exceptions rather than the norm.

The validation matrix makes that boundary explicit: none of the 17 primary studies is coded as directly clinically validated, although several health-oriented studies provide partial clinical or population-specific grounding. Dataset evidence should therefore be interpreted as task- and cohort-level support rather than deployment-level clinical validation.

The dataset catalog should also be read by evidence status. It groups resources into on-body or mobile HAR benchmarks, clinical biosignal or physiological time-series datasets, proxy spatial, pressure, vision, or text-motion resources for cross-modal grounding, study-derived resources rather than reusable independent benchmarks, and non-sensor clinical dialogue datasets for semantic or knowledge-fusion components. In the catalog, “represented” includes direct primary-study use, study-derived resources, synthetic resources, and review-derived contextual resources. Only direct primary-study benchmarks should carry validation weight for generative wearable systems. Several contextual resources enter the catalog through review papers rather than through direct use in primary generative wearable studies. This classification prevents proxy and context datasets from carrying the same evidentiary weight as datasets directly used to validate primary generative wearable systems.

The HM literature shifts the problem rather than eliminating it. Datasets such as VitalDB [34] offer clinically rich biosignals, but they come from specific hospital and perioperative settings that do not transfer cleanly to everyday wearable monitoring. Dialogue datasets used by [9] add semantic richness, yet their connection to wearable-device deployment is indirect. The core limitation is therefore not signal quality alone; it is the mismatch between available data contexts and the deployment settings implied by the papers’ claims.

4.2. Ecological validity, annotation, and multimodal grounding

Most included studies still depend on short-session benchmarks such as UCI-HAR [29], HAPT [35], DSADS [36], and USC-HAD [37]. These datasets support controlled experimentation, but they do not expose models to long-term sensor drift, behavior change, seasonal effects, or adherence failures. That omission is consequential because generative methods are often justified as remedies for data scarcity or imbalance in settings that implicitly assume longitudinal monitoring.

More naturalistic resources such as RealWorld [33], HARTH [38], and HAR70+ [39], along with longer-horizon physiological monitoring evidence described in recent health-monitoring studies [22,25], point in a more relevant direction. Even so, demographically diverse, long-duration, clinically anchored wearable datasets remain scarce across the included studies. Annotation depth is a second bottleneck. Text-grounded approaches such as IMUGPT and TxP depend on datasets or synthetic pipelines that bridge signals with language or structured semantics [7,8], yet most inertial benchmarks still provide flat activity labels rather than temporally aligned semantic annotations.

Table 3
Study-type-stratified synthesis of the 22 included papers. Rows are grouped by study type and domain; the primary-study sequence is mirrored in Table 5.

Study	Type	Domain	Data / modality	Model family	Claimed contribution	Validation setting	Main limitation
Chen et al. [3]	Primary	HAR	Multimodal wearable HAR benchmarks	Hierarchical GAN	Class-conditioned multi-modal synthetic augmentation for wearable HAR.	Five public datasets; accuracy and F1 gains.	No explicit physiological validity or deployment evaluation.
Aleroud et al. [20]	Primary	HAR	Smartphone inertial HAR with postural transitions	GAN + microaggregation	Privacy-aware data alteration with limited accuracy loss.	HAPT smartphone benchmark with privacy-utility comparison.	Narrow benchmark scope and no real-time validation.
Leng et al. [7]	Primary	HAR	IMU, motion, and text-aligned activity descriptions	LLM + text-to-motion pipeline	Reduce annotation effort through language-based cross-modality transfer.	RealWorld, PAMAP2, USC-HAD, HAD-AW, and MyoGym downstream HAR performance.	Depends on source-motion coverage and surrogate motion-to-IMU conversion.
Ye and Wang [6]	Primary	HAR	Cross-user inertial HAR datasets	Conditional Variational Autoencoder (CVAE) + adversarial adaptation	Cross-user domain adaptation with temporal relation attention.	Opportunity, PAMAP2, and DSADS transfer experiments.	Generalization tested on benchmarks only.
Oğul [24]	Primary	HAR	Wrist and chest accelerometer motion streams	Probabilistic symbolic sequence model	Interpretable activity recognition and next-move prediction.	HANDY, HAR/Casale, and HMP benchmarks.	Fixed dictionary limits extensibility.
Ray et al. [8]	Primary	HAR	Pressure maps plus text descriptions	Tokenized pressure model + LLM	Reciprocal text-pressure generation for activity understanding.	Trained on PressLang; evaluated on PresSim, TMD, PmatData, MeX, and PID4TC adaptation.	Semantic gains rely partly on synthetic training data.
Zhang et al. [21]	Primary	HAR	Sensor-based HAR benchmarks	Differentiable augmentation + generative noise	Improve HAR performance through adaptive data augmentation.	Five public datasets across three classifiers.	Limited connection to physiological or clinical validity.
Bacciu et al. [25]	Primary	HM	Longitudinal wearable biometrics + mood events	Neural temporal point processes	Model mood declarations from sparse physiological and event signals.	Eight-week wearable cohort.	Limited clinical/context grounding and no synthetic-data privacy analysis.
Romanelli and Martinelli [28]	Primary	Sensor proxy	Sensor signals with noise characteristics	Autoregressive model + denoising autoencoder	Generate sensor measurements with realistic noise structure.	Ultra Wideband (UWB)/Radio Frequency Identification (RFID) robot-localization signal benchmarks.	Healthcare relevance is application-level, not empirically validated.
Sohn et al. [5]	Primary	HM	ECG-to-PPG biosignal translation	U-Net GAN	Synthetic PPG generation from ECG for Atrial fibrillation (AF) workflows.	VitalDB and long-term ECG resources.	Physiological plausibility only partially examined.
Yang et al. [4]	Primary	HM	Biosignals (ECG and related time series)	Long Short Term Memory (LSTM)-based GAN	Health-data augmentation for scarce biosignal classes.	ECG200, fetal ECG, and mHealth classification tasks.	Emphasis on benchmark accuracy over clinical validation.
Yang et al. [9]	Primary	HM	Wearable outcomes + medical knowledge + dialogue data	LLM + retrieval + knowledge fusion	Diagnosis-oriented dialogue prototype with wearable inputs.	Dialogue benchmarks, self-collected sensor data, and user study.	Limited external clinical validation.
Jeong et al. [22]	Primary	HM	Pediatric wearable physiology and behavior logs	XAI + Wasserstein GAN sampling	Address extreme class imbalance in weight-change prediction.	Seoul cohort plus external Jeju validation with synthetic minority augmentation.	Geographically narrow school cohorts and limited population transfer.
Bukit et al. [10]	Primary	HAR/HM	Federated wearable HAR datasets	Semi-supervised federated learning + LLM filtering	Privacy-oriented distributed learning under label scarcity.	WISDM, PAMAP2, and RealWorld benchmarks.	No direct clinical endpoint validation.
Djemaa et al. [23]	Primary	HAR/HM	Activity-recognition data with rare classes	Controllable GAN + LLM-based classifier	Targeted augmentation for rare and abnormal activities.	WISDM accuracy benchmark.	Clinical claims exceed the validation setting.
Errafik et al. [26]	Primary	HAR/HM	Elderly wearable activity monitoring	Multi-stage augmentation + autoencoding + classification	Improve elderly-monitoring recognition performance.	HARTH and HAR70+ elderly-activity benchmarks.	Limited evidence beyond selected elderly datasets.
Jin et al. [27]	Primary	HAR/HM	Multi-sensor time-series anomaly data	Time-series imaging association network	Abnormal behavior detection with richer temporal associations.	WISDM, HARTH, and LRAD anomaly experiments.	Strong benchmark focus, limited deployment evidence.
Amiri et al. [11]	Review	HM	IoT-based bio and medical informatics	Systematic review	Broad taxonomy of deep learning for medical IoT.	Narrative synthesis of 25 articles.	Scope broader than generative wearable sensing.
Nasarian et al. [12]	Review	HM	Healthcare machine-learning and clinician-AI workflows	Systematic review / framework	Interpretable ML/XAI framing for clinician-AI collaboration.	Narrative synthesis of prior studies.	Not specific to generative wearable validation.
Wang et al. [13]	Review	HM	Smart-glasses digital health literature	Systematic review	Benefits, limitations, and future directions for smart glasses.	Narrative synthesis.	Device-specific scope, not generative wearable datasets.
Kaseris et al. [1]	Survey	HAR/HM	Deep-learning HAR literature	Survey article	Contextualize HAR as a foundation for health monitoring.	Broad literature review.	Not focused on generative data-centric questions.
Li et al. [19]	Framework	HM	Multimodal personal-health streams	Modular AI fusion framework	Closed-loop personalized health-management architecture.	Prototype case study on adaptive relaxation content.	No clinical or multi-dataset validation.

Table 4
Analytical summary of the dataset families represented across the included studies. Illustrative dataset examples are selective rather than exhaustive; the detailed dataset and resource catalog is provided in the Supplementary Material.

Modality class	Representative datasets	Strengths in the included studies	Main constraints for generative wearable research
Motion / inertial	UCI-HAR, PAMAP2, WISDM, HARTH, RealWorld (direct wearable HAR benchmarks)	Established benchmarks, standardized protocols, and strong comparability across augmentation and transfer-learning studies.	Healthy-adult bias, short sessions, limited longitudinal behavior, sparse semantic annotation, and weak clinical anchoring.
Spatial / pressure / vision	UTD-MHAD, PmatData, PresSim, PID4TC (proxy/cross-modal); HumanML3D, Motion-X (text-motion/contextual); PressLang (synthetic study-derived)	Useful for cross-modal grounding, pressure-based activity reasoning, and text-linked motion representations.	Often proxy data for wearables, complex preprocessing, and limited direct linkage to ambulatory health deployment.
Physiological / human-centric	VitalDB (clinical biosignal); ECG200, mHealth (physiological time-series/archive benchmarks); BioBeats, Seoul/Jeju student lifelog, DrHouse sensor data (study-derived); KaMed, DIALMED, MedDG (non-sensor dialogue resources)	Rich biosignals, some longitudinal monitoring, and a stronger connection to health-related tasks.	Narrow collection contexts, privacy constraints, limited multimodal annotation, and variable transfer to consumer-wearable settings.

The same limitation appears in multimodal grounding. Datasets such as UTD-MHAD [40], MeX [41], HumanML3D [42], Motion-X [43], PmatData [44], and PresSim [45] expand cross-modal transfer possibilities, but they were rarely collected to support clinically interpretable generative wearable models. As a result, many papers rely on proxy supervision assembled from separate sources rather than on cohesive wearable-health datasets designed for semantic and health-use grounding.

4.3. Implications for translation

The available datasets support experimentation better than real-world translation. Current benchmarks can demonstrate augmentation gains, cross-modal proof-of-concept systems, and benchmark improvements, but they offer weaker support for claims about generalized health monitoring, physiological realism, or clinically meaningful personalization. Translation claims therefore need more than model innovation: they need more representative cohorts, longer monitoring windows, richer metadata, and tighter alignment between sensing conditions and intended deployment.

5. Discussion and critical analysis

The main finding of this evidence-map review is a mismatch between the gains that papers optimize and the evidence they provide for deployment. Table 5 makes that mismatch visible within the 17 primary studies. The table uses *Y* for direct evaluation, *P* for partial, indirect, context-only, or proxy evidence, and *N* for evidence not explicitly reported. For the Clinical/context grounding column, *P* requires explicit clinical or health-management workflow, target-population, decision-use, or population-specific monitoring context without prospective or external clinical validation; a disease-labelled or physiological benchmark alone is not sufficient.

The descriptive distribution in Fig. 3 shows the imbalance. Distributional fidelity is directly evaluated in 5/17 primary studies (29.4%), partially supported in 5/17 (29.4%), and not explicitly reported in 7/17 (41.2%). Downstream utility is coded *Y* in 16/17 studies (94.1%) and *P* in 1/17 (5.9%). Generalization or transfer is coded *Y* in 3/17 (17.6%), *P* in 9/17 (52.9%), and *N* in 5/17 (29.4%). By contrast, physiological plausibility is never coded *Y*, is coded *P* in only 1/17 (5.9%), and is coded *N* in 16/17 (94.1%). Privacy mechanism/evaluation is coded *Y* in 1/17 (5.9%), *P* in 1/17 (5.9%), and *N* in 15/17 (88.2%). Clinical/context grounding is never coded *Y*, is coded *P* in 4/17 (23.5%), and is coded *N* in 13/17 (76.5%).

The clinical/context *P* cases are heterogeneous: an AF-oriented workflow context, a diagnosis-oriented prototype context, a pediatric weight-management context, and an elderly-monitoring population context are grouped only as partial grounding, not as equivalent clinical validation. Direct clinical validation remains absent from the primary-study matrix. The matrix is a qualitative evidence map rather than a formal risk-of-bias tool or metric-normalized validation framework. The imbalance it shows is the *reality gap*.

5.1. Fusion-specific validation implications

The same validation gap has different implications for different fusion roles. For data synthesis and augmentation, utility gains need to be accompanied by fidelity, minority-class validity, transfer, and privacy checks because generated samples are fused directly into training data. For modality translation, such as motion-to-IMU, pressure-language, or ECG-to-PPG mapping, validation should examine temporal alignment, modality-specific error propagation, and whether the translated signal preserves clinically or biomechanically meaningful relations.

For semantic, language, knowledge, and retrieval fusion, the key risks are source reliability, hallucinated or unsupported explanations, uncertainty communication, and the effect of retrieved knowledge on the downstream decision endpoint. For federated or distributed fusion, benchmark accuracy is insufficient without reporting client heterogeneity, missing-modality behavior, privacy-utility trade-offs, and whether privacy claims are design-level, empirical, or formal. For decision-use prototypes, the validation endpoint should be the stated monitoring, anomaly-detection, diagnosis-oriented, or health-management use case rather than a detached benchmark score.

5.2. Benchmark gains dominate validation

The first part of the reality gap is evaluative. Studies such as HMGAN and Time Series GAN (TS-GAN) report better accuracy or F1 after synthetic augmentation [3,4]; DriveData, Li et al., and related work show similar downstream gains under bounded benchmark conditions [6,21]. These are meaningful results, but they do not by themselves show that generated or transformed signals preserve the physiological, biomechanical, or clinically meaningful relationships required for health deployment. Only a small minority of primary studies move beyond proxy machine-learning outcomes, and even there the evidence remains task-specific rather than general.

5.3. Language models extend semantics, not clinical validation

In the included literature, LLMs function primarily as semantic support components rather than as mature clinical agents. IMUGPT and TxP use language to enrich activity descriptions or connect text with sensor-like outputs [7,8]; Activity Transition Consistency Federated Learning

Table 5
Validation matrix for the primary studies in the review. Rows follow the primary-study sequence used in Table 3. ‘Y‘ = directly evaluated, ‘P‘ = partial, indirect, context-only, or proxy evidence, and ‘N‘ = not explicitly reported. For Clinical/context grounding, ‘P‘ requires explicit clinical or health-management workflow, target-population, decision-use, or population-specific monitoring context without prospective or external clinical validation; a disease-labelled or physiological benchmark alone is not sufficient. Privacy mechanism/evaluation distinguishes empirical evaluation from design-level mechanisms; neither coding implies a formal privacy guarantee.

Primary study	Paper title	Fidelity	Utility	Generalization	Physiology	Privacymech./eval.	Clinical/contextgrounding
Chen et al. [3]	HMGAN: A Hierarchical Multi-Modal Generative Adversarial Network Model for Wearable Human Activity Recognition	Y	Y	P	N	N	N
Aleroud et al. [20]	A privacy-enhanced human activity recognition using GAN & entropy ranking of microaggregated data	P	Y	N	N	Y	N
Leng et al. [7]	IMUGPT 2.0: Language-Based Cross Modality Transfer for Sensor-Based Human Activity Recognition	P	Y	P	N	N	N
Ye and Wang [6]	Deep generative domain adaptation with temporal relation attention mechanism for cross-user activity recognition	N	Y	Y	N	N	N
Oğul [24]	Language of actions: A generative model for activity recognition and next move prediction from motion sensors	N	Y	P	N	N	N
Ray et al. [8]	TxP: Reciprocal Generation of Ground Pressure Dynamics and Activity Descriptions for Improving Human Activity Recognition	P	Y	P	N	N	N
Zhang et al. [21]	Differentiable Prior-Driven Data Augmentation for Sensor-Based Human Activity Recognition	N	Y	Y	N	N	N
Bacciu et al. [25]	Modeling Mood Polarity and Declaration Occurrence by Neural Temporal Point Processes	N	Y	P	N	N	N
Romanelli and Martinelli [28]	Synthetic Sensor Measurement Generation With Noise Learning and Multi-Modal Information	Y	P	N	N	N	N
Sohn et al. [5]	Validation of Electrocardiogram Based Photoplethysmogram Generated Using U-Net Based Generative Adversarial Networks	Y	Y	N	P	N	P
Yang et al. [4]	TS-GAN: Time-series GAN for Sensor-based Health Data Augmentation	Y	Y	N	N	N	N
Yang et al. [9]	DrHouse: An LLM-empowered Diagnostic Reasoning System through Harnessing Outcomes from Sensor Data and Expert Knowledge	N	Y	P	N	N	P
Jeong et al. [22]	Application of Explainable Artificial Intelligence for personalized childhood weight management using IoT data	P	Y	Y	N	N	P
Bukit et al. [10]	Activity transitions for semi-supervised federated learning in sensor-based human activity recognition	N	Y	P	N	P	N
Djemaa et al. [23]	Improved human activity recognition through controllable GAN-Generated synthetic data and large Language models for classification	Y	Y	N	N	N	N
Errafik et al. [26]	An Effective Approach for Wearable Sensor-Based Human Activity Recognition in Elderly Monitoring	P	Y	P	N	N	P
Jin et al. [27]	TIAN: A time series Imaging Association Network for human abnormal behavior detection	N	Y	P	N	N	N

(ATCoFed) uses an LLM within a pseudo-label filtering pipeline [10]; DrHouse combines wearable readings with retrieved medical knowledge in a dialog system [9]. These contributions matter because they extend semantic interfaces, knowledge grounding, and annotation support. They do not, however, constitute evidence that wearable LLM systems are clinically validated diagnostic tools.

5.4. Privacy and deployment claims outpace measurement

Privacy is often cited as a motivation for synthetic data generation, yet only a minority of papers quantify privacy leakage or integrate privacy-oriented learning into the evaluation. Aleroud et al. and Bukit et al. stand out because privacy is part of the technical design rather than

Table 6
Recommended reporting priorities derived from the review.

Observed gap	Pattern in the review	Reporting implication for future studies
Fidelity versus utility	Many studies report synthetic realism or downstream accuracy, but not whether generated or translated signals preserve clinically meaningful relations.	Report both distributional fidelity and task utility, then add problem-specific plausibility checks tied to the target application.
Weak population transfer	Evaluation is often limited to standard public benchmarks or single cohorts.	Report cross-user, cross-device, or cross-cohort tests whenever claims extend beyond a narrow benchmark.
Sparse privacy evidence	Privacy is frequently motivated but rarely quantified; only a minority of studies evaluate privacy mechanisms or privacy utility explicitly.	When privacy is claimed, distinguish privacy-oriented mechanisms, privacy-utility evaluation, empirical leakage or privacy-risk testing, and formal guarantees.
Overstated language-model claims	LLMs often serve semantic or retrieval roles, yet some papers imply broader diagnostic maturity than the evidence supports.	Calibrate claims to the reported evaluation setting and distinguish semantic support from validated clinical decision-making.
Dataset-deployment mismatch	Short-session, healthy-adult benchmarks dominate despite health-monitoring ambitions.	Explain why the chosen dataset fits the deployment claim, and acknowledge where benchmark limitations bound external validity.
Underreported fusion architecture	Fusion roles are often implicit across generated data, translated modalities, language grounding, retrieval, federated learning, or decision-use interfaces.	Report the fusion level, fused evidence streams, synchronization/alignment assumptions, missing-modality handling, uncertainty propagation, and decision endpoint.
Clinical-readiness gap	Health-oriented papers rarely specify the operational pathway from model output to a safe clinical or monitoring decision.	State intended use, target population, human workflow, safety and failure modes, clinical endpoint, and regulatory status when applicable.

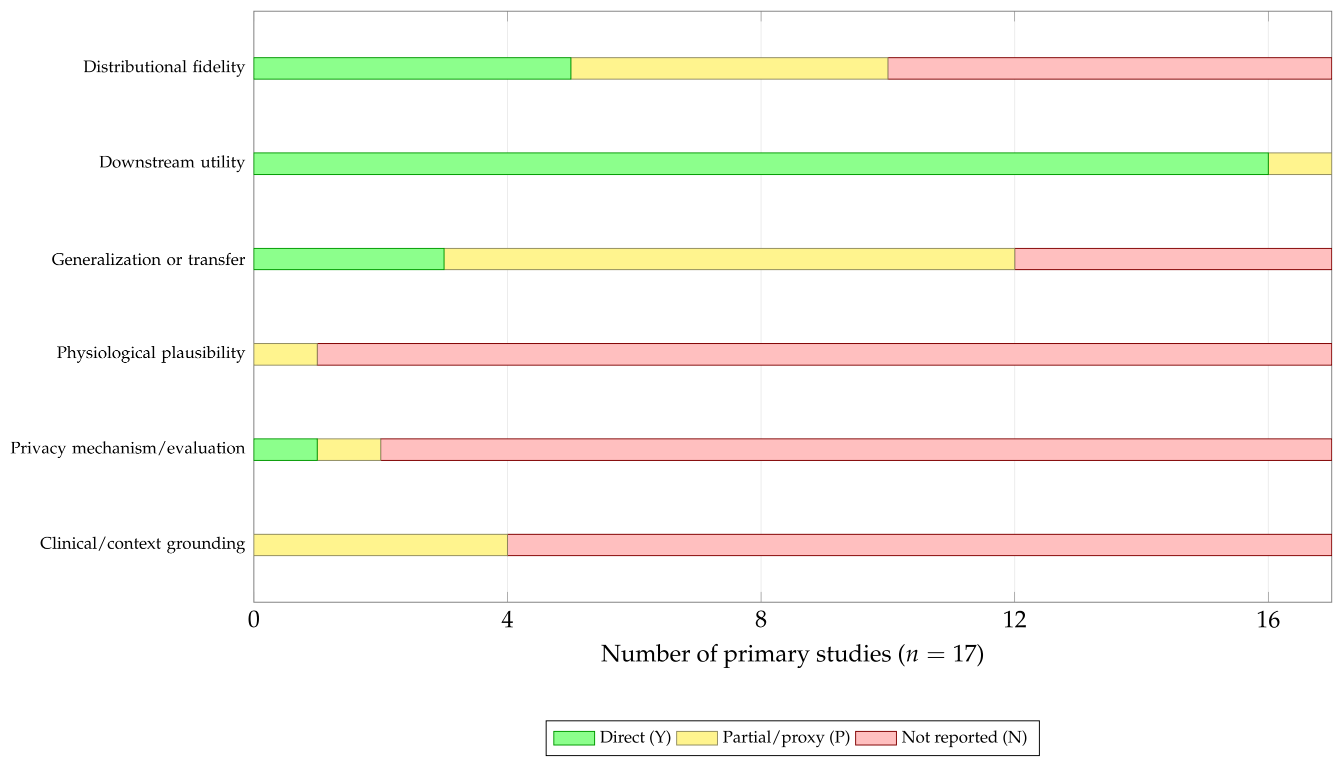

Fig. 3. Descriptive validation-gap summary for the 17 primary studies. Counts are derived from Table 5; they summarize evidence-map coding rather than pooled effect sizes or formal study-quality scores.

a narrative motivation [10,20]. In this evidence map, privacy coding should be read conservatively: *Y* indicates empirical privacy evaluation or privacy-utility evidence in the source study, whereas *P* indicates a privacy-oriented mechanism or design without preserved evidence of empirical leakage testing or formal guarantees. Future reporting should distinguish privacy-motivated design, privacy-utility evaluation, empirical leakage or privacy-risk testing, and formal privacy guarantees.

The same pattern holds for deployment and clinical validation. Computational feasibility is frequently asserted, but latency, energy consumption, and memory behavior on wearable hardware are rarely measured formally. Clinical deployment claims also require more than a health-related dataset or decision-use narrative: they require intended-use definition, target-population grounding, longitudinal monitoring conditions, human-workflow fit, safety considerations, and validation against clinically meaningful endpoints. The reviewed primary studies mostly stop before that level of evidence.

A related limitation is that this evidence map does not audit the full governance and safety conditions needed for clinical or consumer deployment. The preserved corpus coding does not systematically evaluate consent procedures, data-governance arrangements, legal compliance, medical-device status, human-factors safety, clinical oversight, or post-deployment monitoring. These issues therefore remain outside the evidentiary support of the present synthesis, even when individual source studies use health-oriented language.

5.5. Interpretive boundary of the review

The broader reviews and the framework paper support this interpretation but do not resolve it empirically. They highlight multimodal integration, trustworthiness, privacy, and dataset quality, while the framework paper adds a systems perspective [1,11–13,19]. Together, they explain why the gaps matter without substituting for multi-dataset empirical validation.

The review's own scope sets a further boundary on interpretation. As it focuses on Scopus-indexed Q1/Q2 journal literature anchored to a frozen source-of-record retrieval archived on 3 December 2025, it does not claim full recall across conferences, lower-quartile journals, adjacent interdisciplinary venues, post-snapshot publications, or later-indexed records. Its value is therefore selective and interpretive: it clarifies what the documented included corpus currently validates, and what it still leaves under-evaluated.

5.6. Recommendations for future work

The recommendations in Table 6 follow directly from this evidence pattern. Future work should report layered validation instead of benchmark accuracy alone, tie dataset choice to the intended deployment context, quantify privacy when privacy is claimed, and calibrate the language used for LLM-enabled health systems. Each paper should also make the dataset, evaluation metric, information-fusion role, and claimed real-world or clinical use case correspond explicitly, rather than leaving those elements separated across method, results, and discussion sections.

6. Conclusion

This focused evidence-map review covers a frozen corpus of 22 Scopus-indexed Q1/Q2 journal papers drawn from the targeted 2023–2025 period and anchored to the archived 3 December 2025 Scopus retrieval. Three conclusions follow from the documented included corpus.

First, generative wearable information fusion is still driven primarily by data-centric use cases. GANs are the most frequent family for augmentation, minority-class balancing, and biosignal synthesis, while other model families appear more selectively in domain adaptation, representation learning, or symbolic sequence modeling. LLMs function mainly as semantic, retrieval, or annotation-support components rather than as clinically validated systems.

Second, datasets remain a central constraint on credible progress. The dominant benchmarks are valuable for controlled comparison, but many are short-duration, laboratory-oriented, and demographically narrow. The reviewed literature can therefore demonstrate benchmark gains without demonstrating reliable performance under the longitudinal, diverse, and clinically grounded conditions that future wearable-health deployment will require.

Third, the primary studies reveal a corpus-bounded reality gap between what is optimized and what is validated. Downstream utility is reported far more consistently than physiological plausibility, privacy mechanisms or evaluation, or clinical/context grounding. Generalization evidence is present in several studies, but it is usually benchmark-bounded rather than evidence of deployment-level transfer. The reported technical gains remain meaningful, but the evidentiary basis for deployment claims is still limited.

This review is intentionally selective rather than exhaustive. The documented Scopus-indexed Q1/Q2 journal corpus therefore supports narrower conclusions than the deployment-oriented language in some included papers implies. Stronger claims will require richer longitudinal datasets, layered validation, explicit privacy reporting, and clearer links between dataset design, evaluation practice, information-fusion role, and the real-world use cases invoked by the literature.

CRediT authorship contribution statement

Ikramullah Khan: Writing – review & editing, Writing – original draft, Methodology, Investigation; **Jaime A. Martins:** Writing – review & editing, Validation, Supervision, Project administration; **Pedro J.S. Cardoso:** Writing - review & editing, Validation, Supervision, Project administration.

Ethics approval

This review used only published literature and did not generate new human-participant, animal, or clinical data; no new ethics approval was required for the review itself.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI ChatGPT/Codex to support revision planning, editorial checking, language refinement, and LaTeX preparation. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No new primary datasets were generated for this review. The screening count summary, study-type classification, final-corpus traceability table, extraction schema, validation dimensions, and dataset and resource catalog supporting the findings are provided in the article and Supplementary Material. Selection evidence is documented at the aggregate and final-corpus level rather than as a complete record-level audit trail. All source studies used in the synthesis are cited in the reference list.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at 10.1016/j.inffus.2026.104592

Supplementary Material documents the archived Scopus search expression, screening count summary, study-type classification, final-corpus traceability table, extraction and validation schema, and the dataset and resource catalog underlying this focused evidence-map review.

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